



SOFTWARE DEVELOPER TRAINEE – INTERVIEW ASSIGNMENT

Assignment

Build a simple application that monitors the health and performance of a PostgreSQL database and provides basic insights using AI.

Application Type

The application can be implemented as a Web application or a Desktop application. A graphical user interface is optional. Console output or simple reports are acceptable.

Functional Requirements

- Connect to a PostgreSQL database (local or test setup).
- Periodically collect basic metrics such as CPU usage, memory usage, active connections, and slow queries.
- Store collected data locally (files like JSON/CSV are sufficient).
- Allow inspection of current values and simple historical trends.

AI Usage

Use of AI is encouraged but not mandatory. AI can be used to analyze metrics, detect unusual trends, generate human-readable explanations, and suggest basic investigation or optimization steps.

Technical Constraints

- Any programming language or framework may be used.
- PostgreSQL should run locally or in a test environment.
- No production systems are required.

Deliverables

- Runnable application source code.
- Public Git repository (GitHub, GitLab, etc.).
- README with setup instructions, architecture overview, features implemented, and explanation of AI usage.

Evaluation Criteria

Code clarity, correctness, simplicity, practical AI usage, and documentation quality will be evaluated.

Bonus (Optional)

Simple charts, monitoring alerts, configurable intervals, or additional useful features.

Required Data

- A link to the public repository

Expectations

- Allow NAPA to review the code

Additional Notes

During the interview, we will want you to walk us through the process of building the features. We are mainly interested in the thought process behind the design of the features and how you used AI to implement them.